RAGHAVENDER REDDY GAJJALA

SOFTWARE DEVELOPMENT ENGINEER | APPLIED AI/ML ENGINEER

Hyderabad, Telangana, India • +91 8328486501 • raghavender8761reddy@gmail.com

PROFESSIONAL SUMMARY

Software Development Engineer I and AI/ML graduate with experience building backend services, distributed systems, cloud-based data platforms, machine learning solutions and multi-agent applications at Amazon and PwC India. Hands-on experience across Python, REST APIs, AWS and Azure, deep learning, computer vision, NLP, LLM integrations, agent orchestration and cloud security. Combines software engineering fundamentals with applied AI/ML and production-focused system development.

TECHNICAL SKILLS

Programming: Python, Java, C++, Go, SQL, JavaScript

Backend & System Design: Microservices, REST APIs, Distributed Systems, HLD, LLD, SOLID, OOP, Concurrency, System Design, Software Architecture

Machine Learning & Deep Learning: PyTorch, TensorFlow, Keras, Scikit-learn, Neural Networks, CNNs, Transformers, Semantic Segmentation, Feature Engineering, Hyperparameter Tuning, Model Evaluation

Generative & Agentic AI: LLMs, RAG, Fine-tuning, Prompt Engineering, Embeddings, Tool Calling, Multi-Agent Systems, LangChain, LangGraph, MCP, Anthropic Claude

Data Engineering & Analytics: Apache Spark, PySpark, AWS Glue, ETL, Pandas, NumPy, Matplotlib, Seaborn, Statistical Analysis, A/B Testing, Parallel Processing

Databases & Cloud / DevOps: SQL, NoSQL, DynamoDB, Amazon S3, Neo4j, FAISS, Pinecone, OpenSearch, AWS, Azure, GCP BigQuery, SageMaker, Docker, Kubernetes, Terraform, AWS CDK, Jenkins, CI/CD, Git, MLOps

PROFESSIONAL EXPERIENCE

AMAZON | Software Development Engineer I

Jun 2025 – Present | Hyderabad, India

- Designed and developed a multi-agent platform consolidating information from 11 dashboards and 27 data sources, providing program slotting managers with answers and actionable slotting insights for warehouse products.
- Supported a warehouse slotting program associated with an estimated $8.2M in annual value; the platform saves approximately 150 hours of slotting-manager effort per quarter.
- Fine-tuned deep learning models using Amazon SageMaker, PyTorch and TensorFlow, improving classification accuracy from 67% to 89%; evaluated performance using precision, log loss and RMSE.
- Developed local Model Context Protocol (MCP) integrations for secure tool calling and operational workflow automation, reducing manual effort by 20%; implemented guardrails for unsafe tool interactions, hallucinations and system vulnerabilities.
- Optimized Spark ETL pipelines by pruning downstream schemas, reducing the resulting data footprint from megabytes to kilobytes.
- Engineered backend APIs and data-processing components for a separate $7.1M program, improving data ingestion and processing accuracy from 78% to 91%.

PwC INDIA | Cloud Security Engineer – Intern & Apprentice

Feb 2024 – Jun 2025 | Bengaluru, India

- Developed multi-cloud security capabilities using Anthropic Claude to analyze security patterns and query graph relationships stored in Neo4j.
- Built AWS Glue data-lake and ETL pipelines, implementing 100+ PySpark scripts for parallel data processing.
- Supported migration of legacy applications to cloud-native microservices using AWS, Kubernetes and Terraform.
- Developed LangGraph- and MCP-based workflows for multi-agent planning, reasoning and tool calling; built foundation-model workflows for Text-to-SQL and document analysis, saving approximately six hours of manual effort daily.
- Automated infrastructure provisioning and deployment workflows using Jenkins and Terraform.
- Resume-reported outcomes include a 22% improvement in early risk detection and a 30% reduction in manual workload through autonomous security and operational checks.

SELECTED PROJECTS

Multi-Cloud Agentic Security Intelligence & MCP Automation Platform

Technologies: Python, Flask, LangGraph, MCP, LLMs, Neo4j, AWS, Azure, AWS Glue, S3, DynamoDB, PySpark, Terraform, Kubernetes, Jenkins

- Aggregated AWS and Azure logs and mapped cloud-resource relationships in Neo4j; developed data-ingestion pipelines with AWS Glue, S3, DynamoDB and 100+ PySpark scripts.
- Designed agent workflows for security analysis against CIS benchmarks, HIPAA and PII controls; developed autonomous security checks, LangGraph scanning and Text-to-SQL workflows.
- Architected an MCP integration layer for tool discovery, context exchange and controlled tool invocation; supported cloud-native deployment using Kubernetes, Terraform and Jenkins.

Full-Stack Security Intelligence Dashboard

Technologies: React.js, Python, Flask, REST APIs, Neo4j, SQL, AWS

- Developed a React.js frontend and Python Flask backend with REST API integrations for security-data retrieval and processing.

- Used Neo4j graph modeling to represent relationships among cloud resources, systems, events and security findings; organized the application into reusable frontend components and modular backend services.

Deep Learning-Based Polyp Identification & Segmentation

Technologies: Python, Flask, DeepLabV3+, ResNet, Deep Learning, Computer Vision

- Developed a semantic-segmentation solution for identifying and delineating colorectal polyps in colonoscopy images.
- Implemented DeepLabV3+ with a ResNet encoder, using deep visual features and atrous convolutions to retain spatial detail and segment irregular polyp regions.

Transformer-Based Abstractive Text Summarization

Technologies: Python, PyTorch, Transformers, SimpleT5, PyWebIO

- Configured and fine-tuned T5-based models with SimpleT5 to generate concise abstractive summaries of lengthy articles.
- Built a PyWebIO interface for article submission and summary display; used Python for text preparation and model-processing workflows.

Multi-Agent Warehouse Slotting Intelligence Platform

Technologies: Multi-agent systems, data integration, operational intelligence

- Developed a multi-agent platform consolidating information from 11 dashboards and 27 data sources to provide program slotting managers with answers and actionable warehouse-product slotting insights.
- Supported a program associated with an estimated $8.2M in annual value and approximately 150 hours of slotting-manager effort saved per quarter.

EDUCATION

CVR College of Engineering, Hyderabad | B.Tech, Computer Science and Engineering – Artificial Intelligence and Machine Learning | May 2024 | GPA: 9.67/10

Narayana IIT Academy, Hyderabad | Higher Secondary Education – MPC | June 2020 | 98.4%

Narayana E-Techno School, Hyderabad | Secondary School Education | April 2018 | GPA: 9.3/10

CERTIFICATIONS

- Microsoft Certified: Azure AI Fundamentals (AI-900) – 2025
- AWS Certified Cloud Practitioner (AWS-CLF-C02) – 2024
- Data Analytics with Python – NPTEL – 2023
- Big Data Analytics – NPTEL – 2022
- Artificial Intelligence – Wipro Future Skills – 2022
- Cloud Computing – Wipro Future Skills – 2022

PUBLICATION & ACHIEVEMENTS

Publication: Feature-Based Human Face Generation System Using Diffusion Model for Criminals Identification – March 2024. Applied diffusion-model techniques to generate high-fidelity human faces while preserving facial features relevant to the research objective; considered ethical use and privacy implications.

- University Topper – B.Tech, 2020–2024
- Runner-up – Smart India Hackathon 2022
- Web Developer – International Conference MIWAI 2023
- G1 – NASA Space Settlement Design Project
- 1st Prize – Paper Presentation, CYPHER 2022
- 1st Prize – Paper Presentation, CIENCIA 2022
- Rank 889 – Unified Cyber Olympiad